

Phlorotannin assay

Folin Ciocalteu

Using the FLUOstar OPTIMA plate photometer

Sample preparation and extraction

Perform all work steps in the dark and cold!

Freeze-dry samples 24 h

Extraction

- Pulverize samples in FastPrep, 60 sec (3x20 sec)
(Program 1: P1 S 4.0, P1 TH MP, P1 T 20, i.e. Speed: 4.0 m / s, Attachment: MP: 24x2 (White Tower), Time: 20)
- Approximately weigh in 10 mg algae powder per sample & transfer to an Eppi
- Add 1 mL of 70% acetone in each Eppi
- Vortex Eppi
- Extract soluble phlorotannins from algae samples (24 h, 4 °C (in the climatic chamber), darkened with tin foil, on a shaking table)

Phlorotanninbestimmung

- 1) Vortex the algae samples well
- 2) Centrifuge algae samples (10 min, 2500 rpm)
- 3) Remove 50 µL of supernatant and transfer to a new Eppi
(Note: The phlorotannin determination is carried out in triplicate, ie 3 x 50 µL of supernatant per sample of algae are removed!)
- 4) add 250 µL dH₂O
- 5) Add 200 µL of 20% Na₂CO₃
- 6) Add 100 µL of 2N Folin-Ciocalteu reagent
- 7) Incubate algae samples (45 min, RT, darkened with aluminum foil)
- 8) Centrifuge algae samples (3 min, 5000 rpm)
- 9) Pipette 250 µL supernatant per well into the microplate (96 wells) (Note: put plate on styrofoam to avoid temperature discrepancies)
- 10) Measure absorbance at 730 nm

(Note: Absorbance measurements are possible in the following range: -0.100-3.000 OD, with reproducibility 0.010 OD for 0-2 OD range & 0.030 OD for 2-3 OD range!)

Operation of the plate photometer:

- 11) Turn on the computer
- 12) Log in under the user name Fluostar (password is not required)
- 13) Open software FLUOstar OPTIMA
- 14) Test Setup → Set up Test Protocol (Basic Parameters) or select a previously set up Test Protocol
Note: The Basic Parameters of the test protocol for the phlorotannin determination is called A_NORA_PHLORO & as follows:
- 15) Test Setup → Test Protocol (Layout): Specify the wells to be measured according to the assignment of the microplate (**Note: With Sample (...) you can enter samples to be measured & with Empty (...) you can delete them!**) → OK
- 16) Plate Out → Place the microplate in the plate photometer → Plate In
- 17) Measure → Measure → Select Test Protocol KRISTINA → OK
- 18) Apply Gain Adjustment → Gain Adjustment
- 19) Start test run
(Note: Absorption measurements will be carried out! After that the software FLUOstar OPTIMA will close!)
- 20) Open software FLUOstar OPTIMA again
- 21) Evaluation Part → Select appropriate Excel file → Copy and save the absorption values from the worksheet Raw Data or Evaluation to a new Excel file
(Note: In the Raw Data worksheet, the absorption values are "disorderly" and in the evaluation sheet they are according to the Assignment of the microplate before!)

Calibration with phloroglucinol:

- 22) Prepare sufficient (10 mL) of 1 g/L phloroglucinol standard solution
(0.01 g/mL) 10 mg in 10 mL Ethanol
- 23) Prepare 10 dilution steps of the phloroglucinol standard solution in Eppis (0-300 µg/mL, see Preparation of dilution steps - phloroglucinol)
- 24) Remove 50 µL of supernatant and transfer to a new Eppi
(Note: The determination of phlorotannin is carried out in duplicate, that is to say two 50 µL supernatants per dilution stage are removed!)
- 25) add 250 µL dH₂O
- 26) Add 200 µL of 20% Na₂CO₃ → **10g in 50 mL ddH₂O;**
- 27) Add 100 µL of 2N Folin-Ciocalteu reagent
- 28) Incubate dilution stages (45 min, RT, darkened with aluminum foil)
- 29) Centrifuging dilution stages (3 min, 5000 rpm)
- 30) Pipette 250 µL of supernatant per well into the microplate (96 wells)

31) Measure absorbance at 730 nm

(Note: Absorbance measurements are possible in the following range: -0.100-3.000 OD, with reproducibility 0.010 OD for 0-2 OD range & 0.030 OD for 2-3 OD range!)

32) Apply measured absorbances to corresponding phloroglucinol concentrations ($\mu\text{g/mL}$) (scatterplot)

(note: blank (= absorbance of the 0 $\mu\text{g/mL}$ dilution step) should be subtracted from all absorbance values before!)

33) Drag the linear trend line through the points, display the straight line equation & the coefficient of determination